

Benchmark #7

1 MeV H on Fe

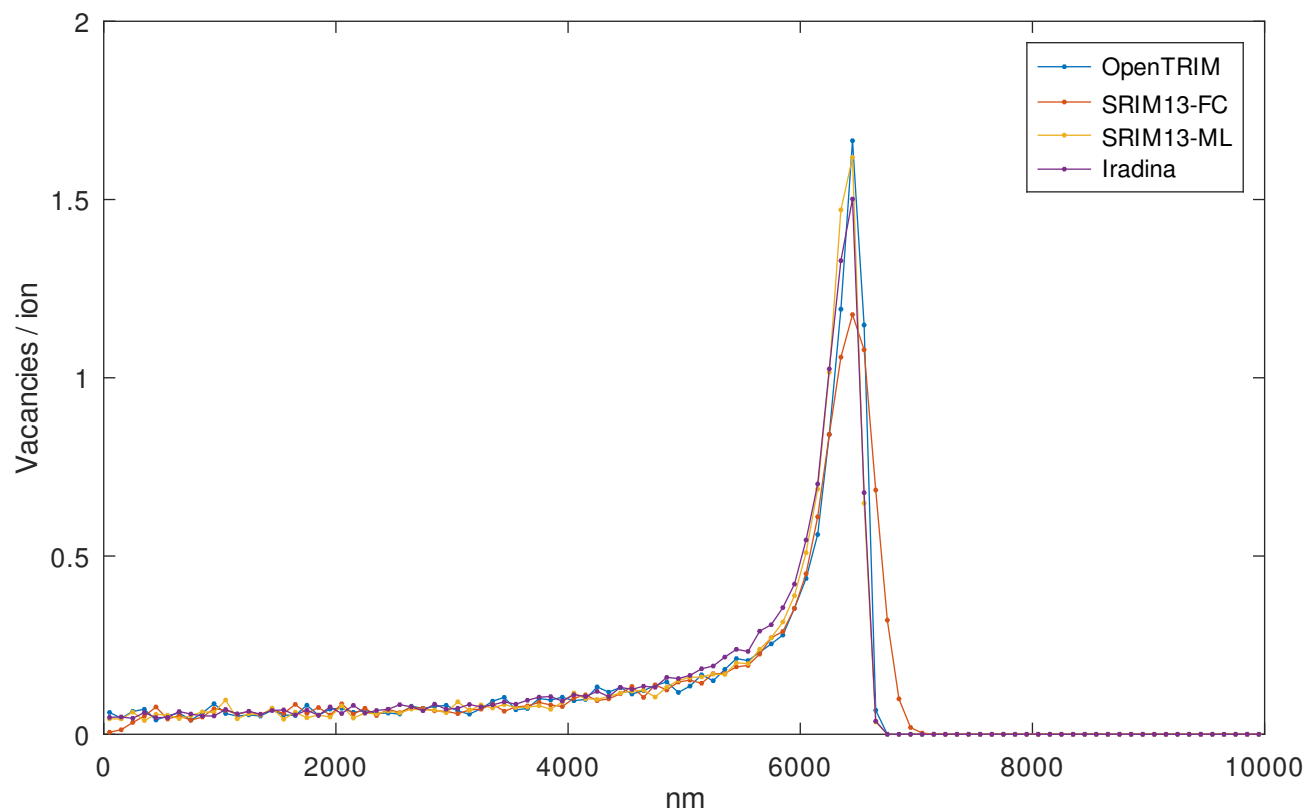
Ion energy $E_0 = 1\text{e}+06$ eV

Target depth = 10000 nm

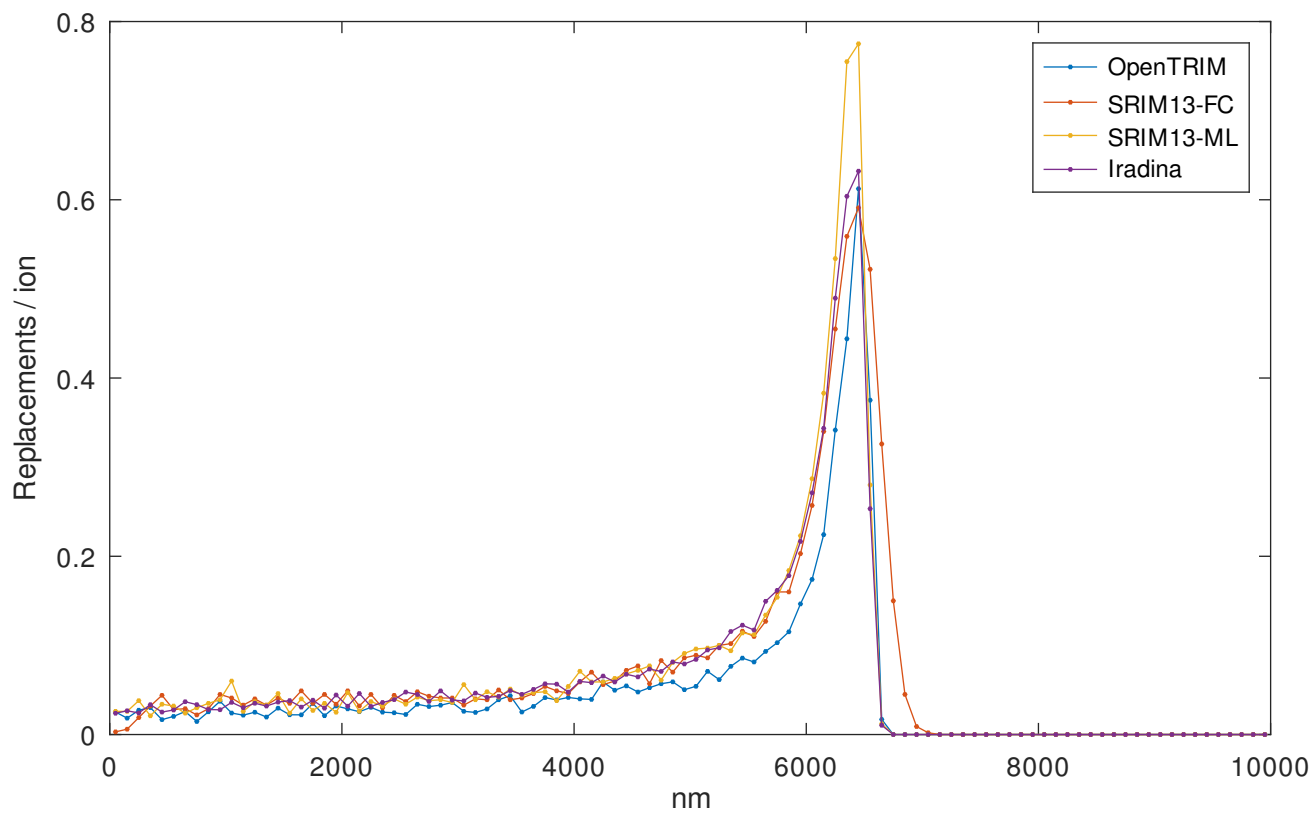
Summary Table

Quantity	OpenTRIM	SRIM13-FC	SRIM13-ML	Iradina
$V(\text{Fe})$	12	12.2	11.9	12.5
$R(\text{tot})$	4.71	6.69	6.53	6.14
$I(\text{H})$	1	1	1	1
$\Delta E_e(\text{H})/E_0$	0.998	0.998	0.998	0
$\Delta E_e(r)/E_0$	9.19e-05	0.000104	0.000102	0
$\Delta E_e(\text{tot})/E_0$	0.998	0.998	0.998	0.998
$\Delta E_n(\text{H})/E_0$	0.000781	0.000654	0.000915	0
$\Delta E_n(r)/E_0$	0.000971	0.00154	0.00178	0
$\Delta E_n(\text{tot})/E_0$	0.00175	0.00219	0.00269	0.00183
$1 - (\Delta E_e + \Delta E_n)/E_0$	1.79e-07	-0.00055	-0.000704	0

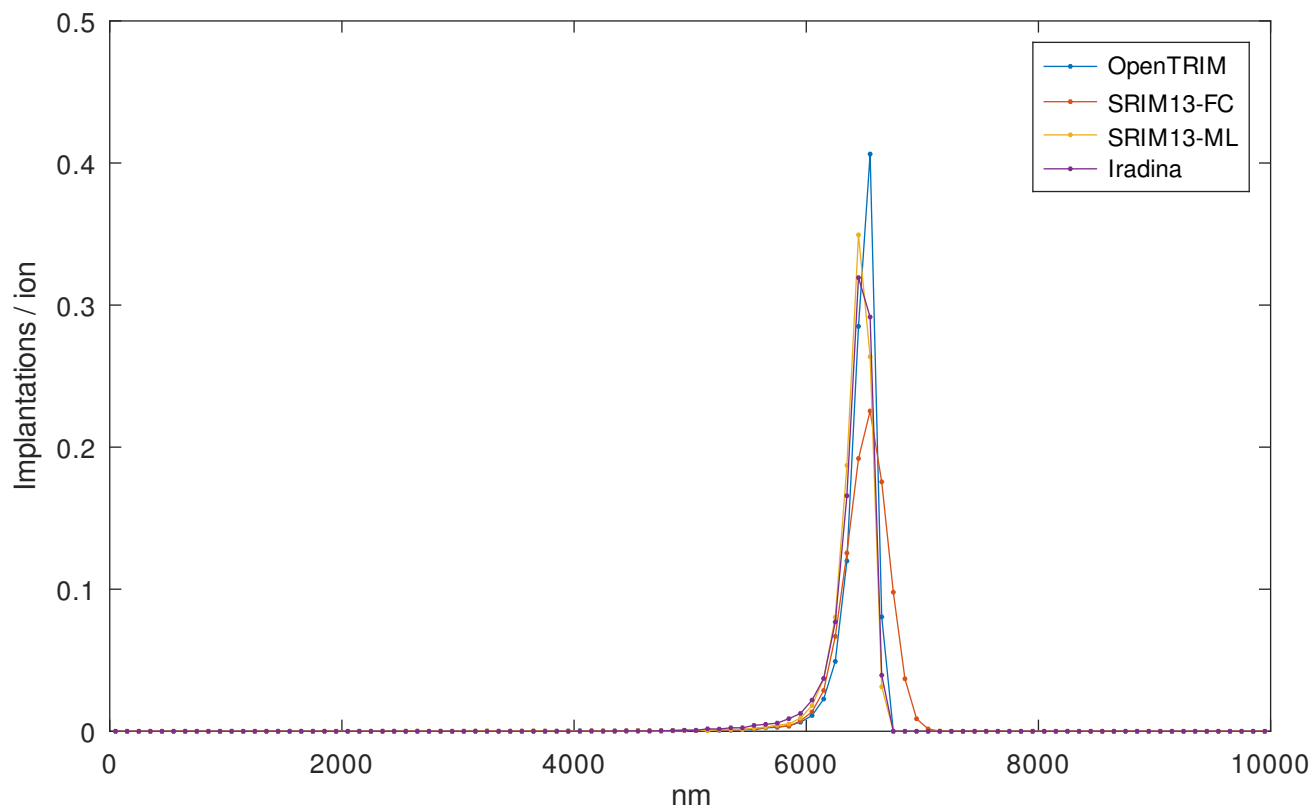
Vacancies of Fe in Fe



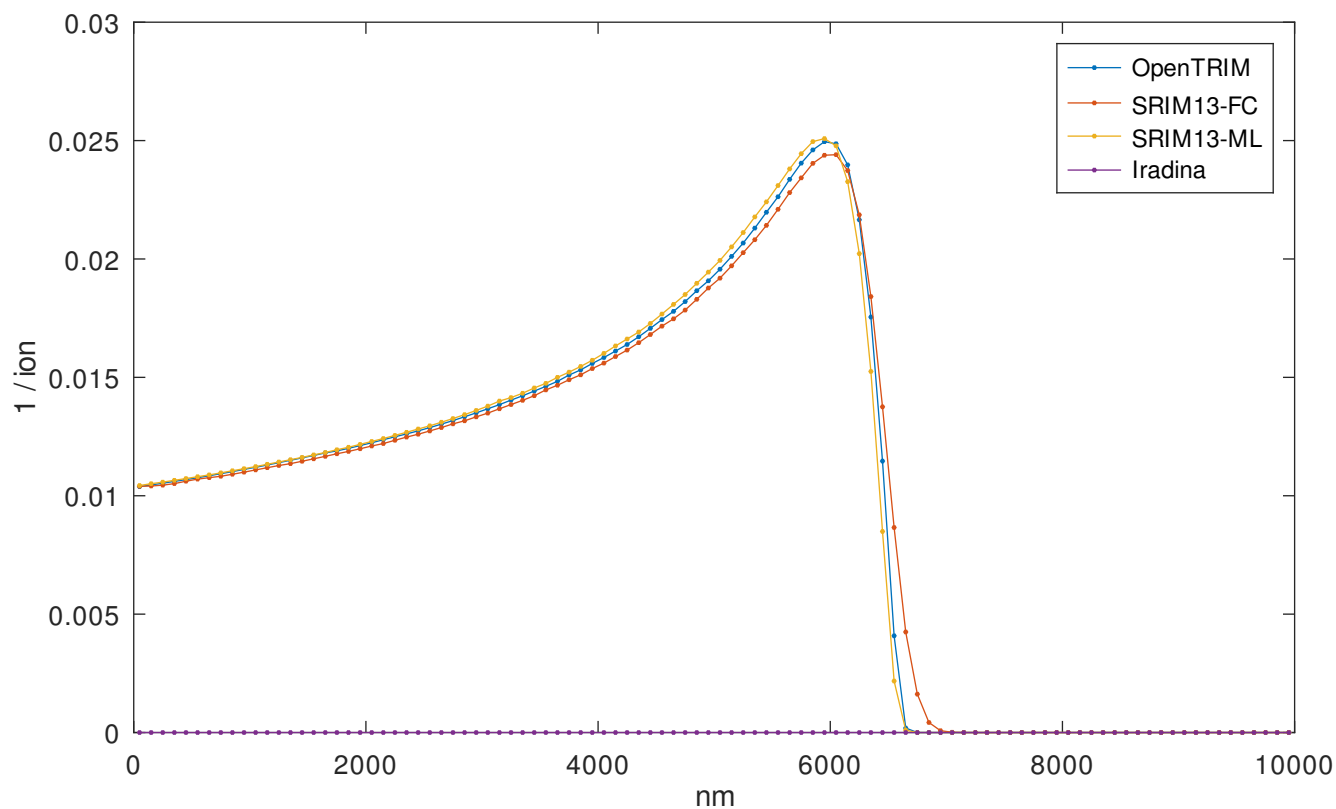
Replacements



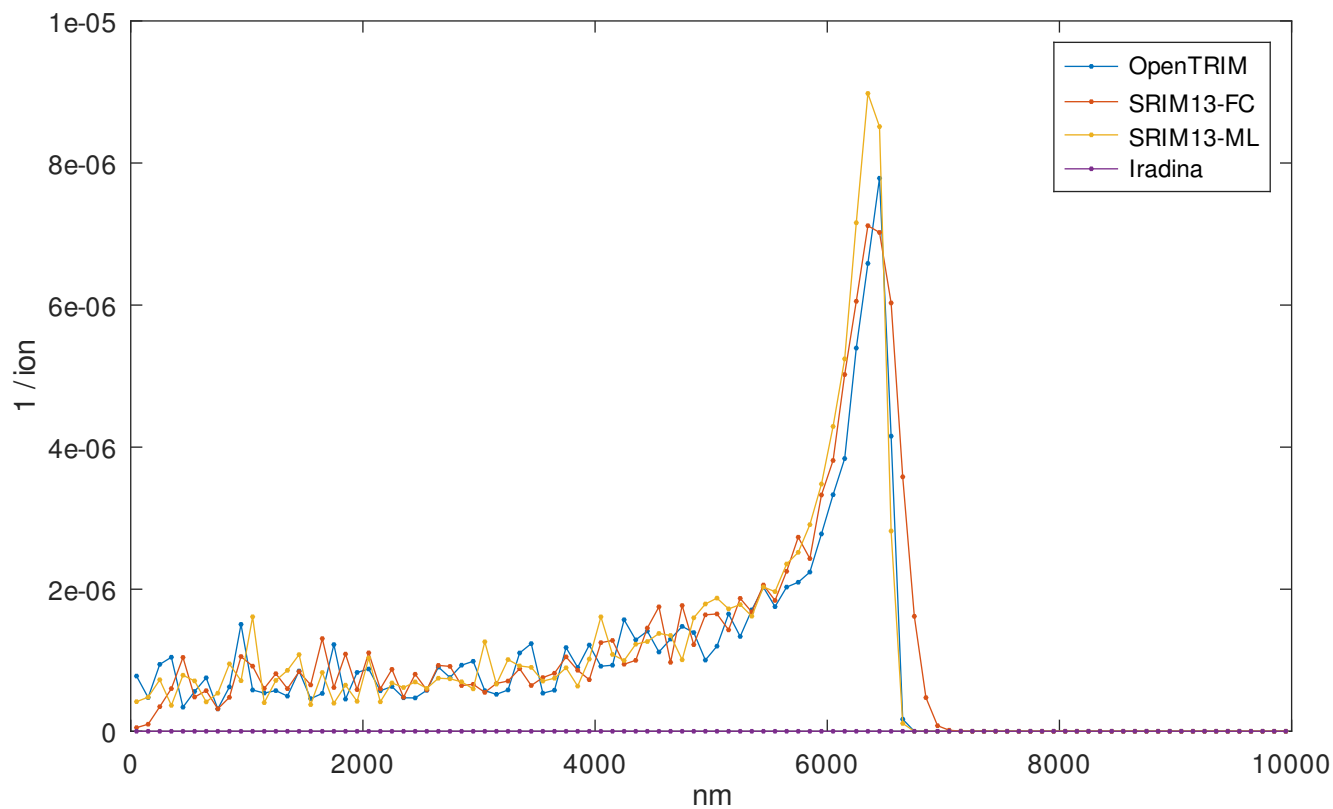
Implanted H ion



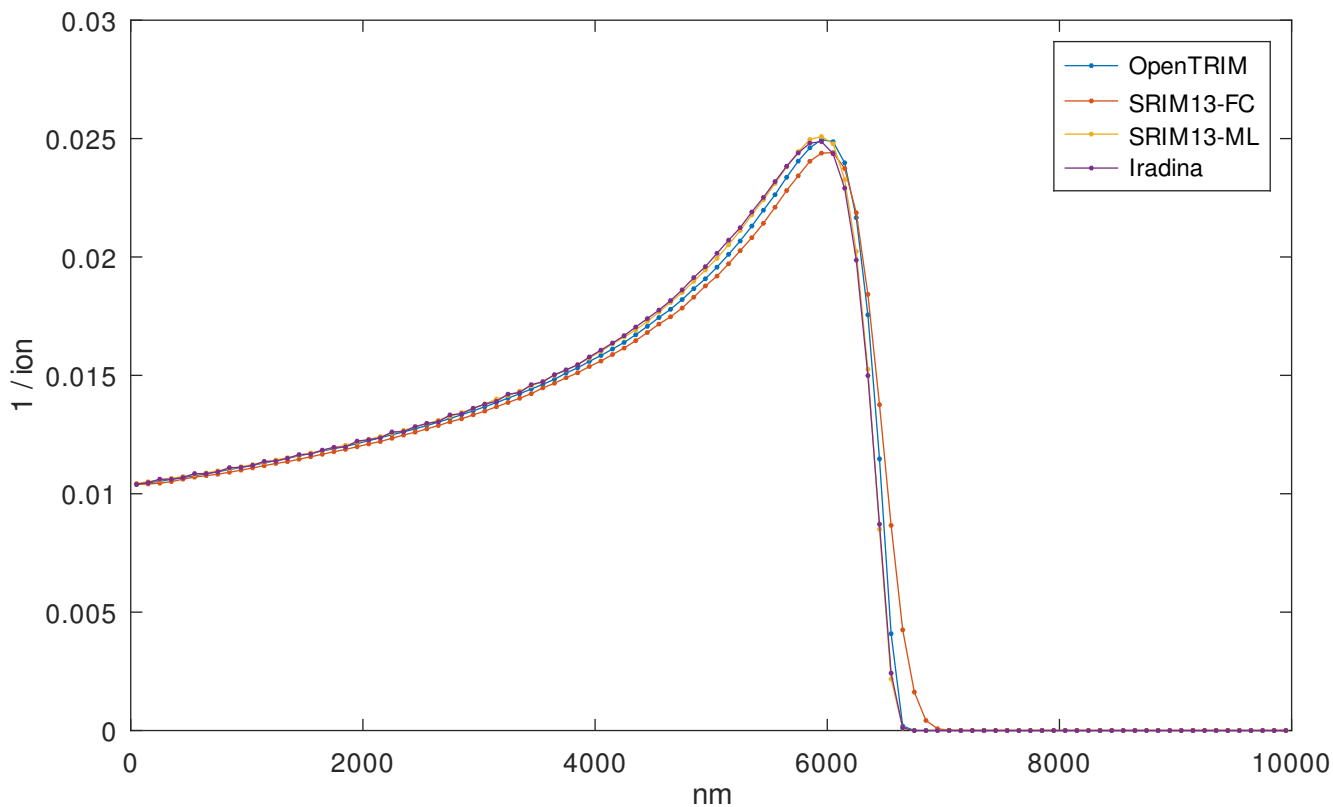
Electronic energy loss fraction $\Delta E_e/E_0$ by H ion



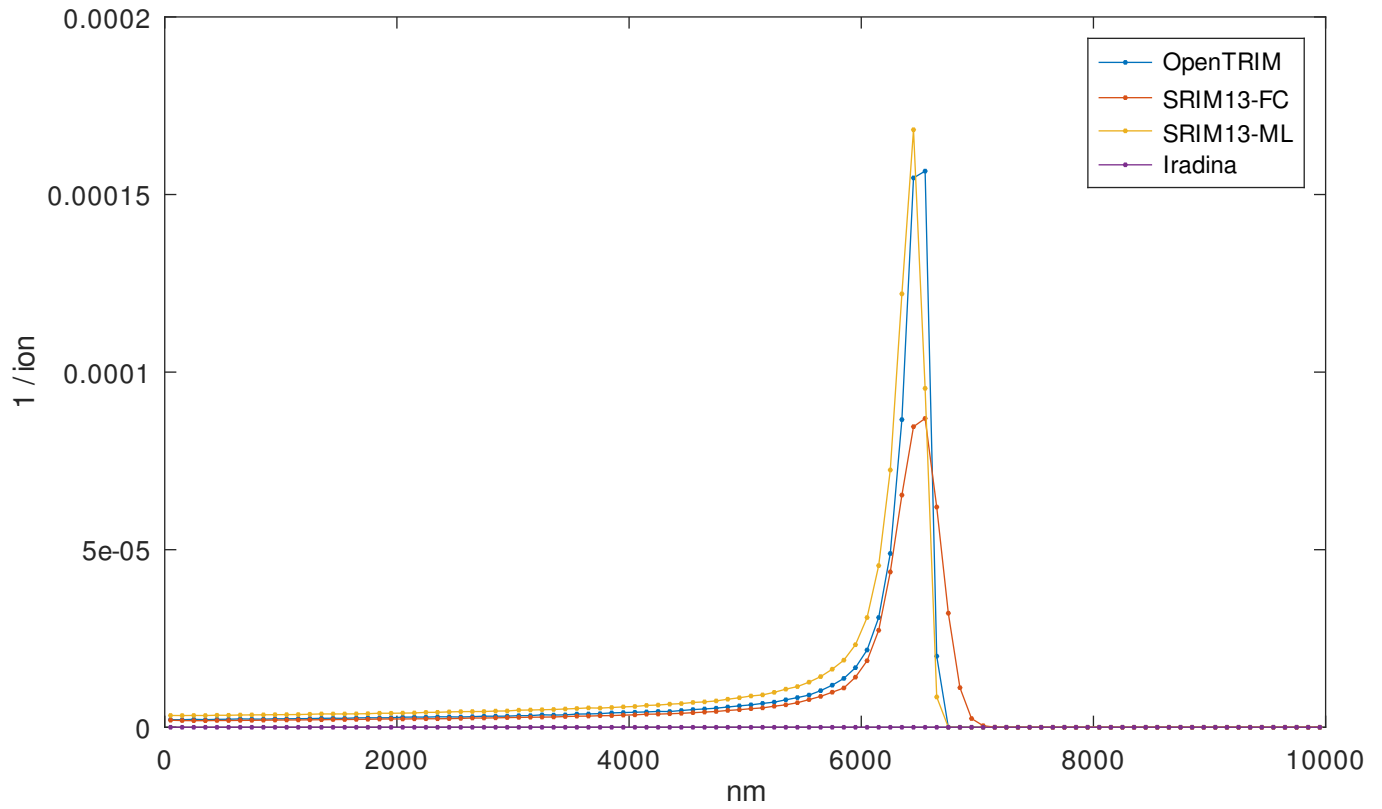
Electronic energy loss fraction $\Delta E_e/E_0$ by recoils



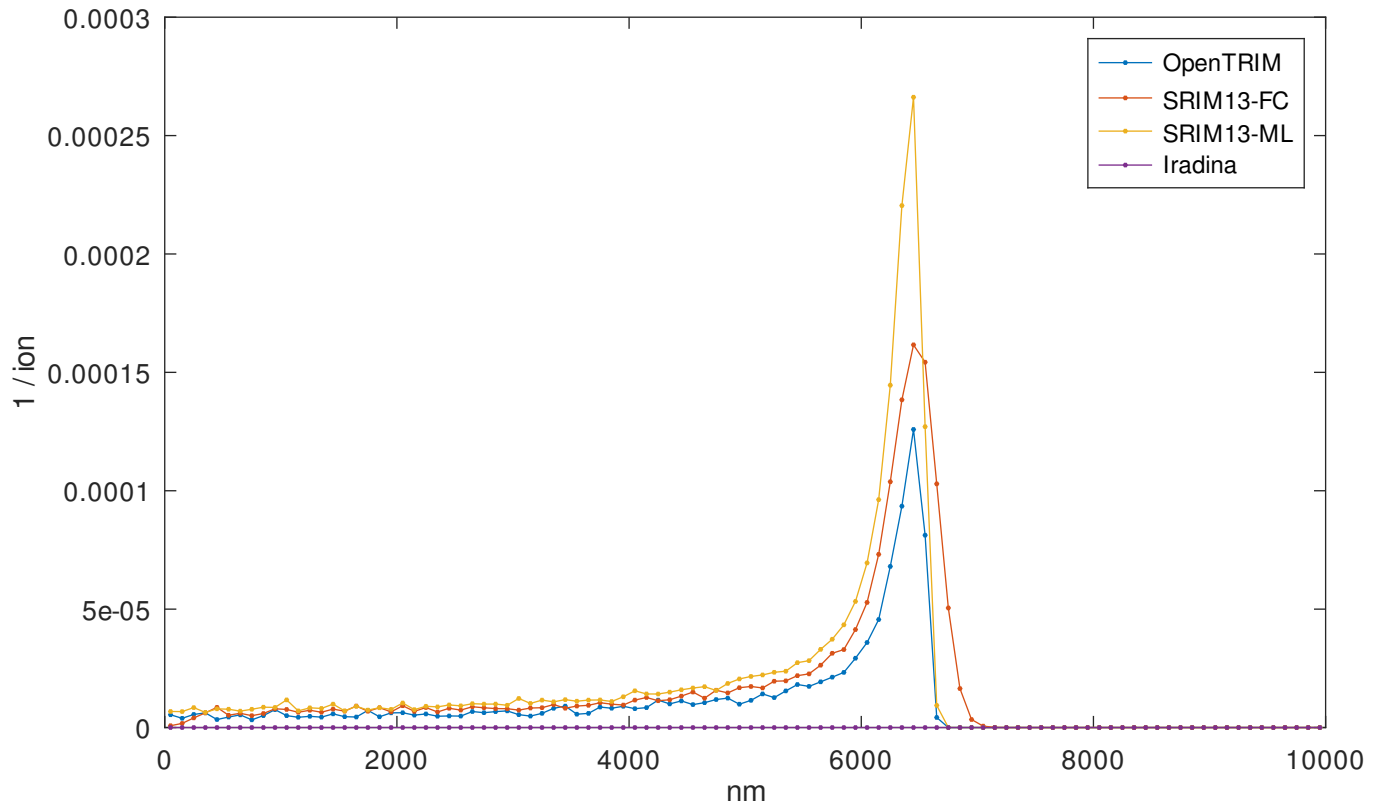
Total Electronic energy loss fraction $\Delta E_e/E_0$



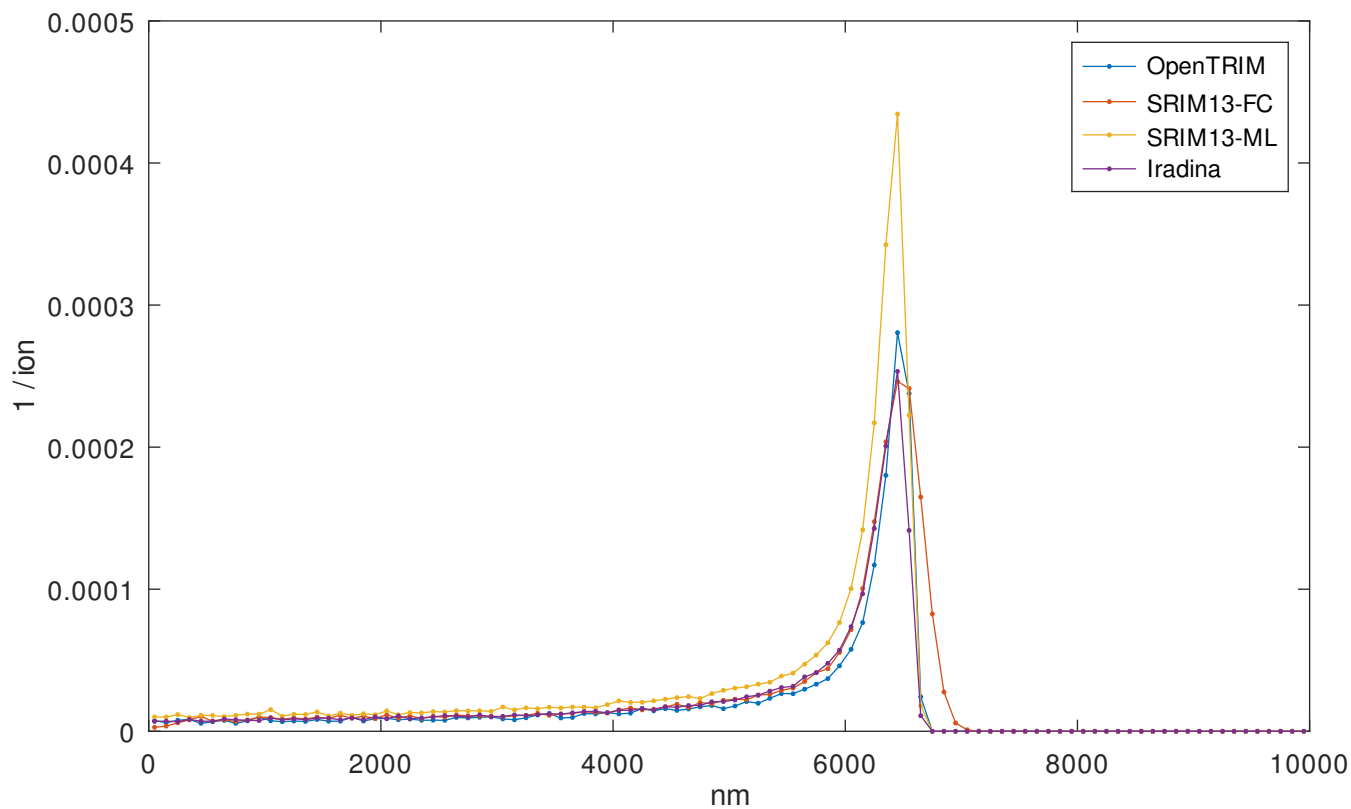
Nuclear energy loss fraction $\Delta E_n/E_0$ by H ion



Nuclear energy loss fraction $\Delta E_n/E_0$ by recoils



Total Nuclear energy loss fraction $\Delta E_n/E_0$



Total fractional energy loss $(\Delta E_e + \Delta E_n)/E_0$

